Improving LLM Final Representations with Inter-Layer Geometry

Tom Ulanovski*
Blavatnik School of Computer Science
Tel Aviv University
tomulanovski@mail.tau.ac.il

Eyal Blyachman*
Tel Aviv University
blyachman1@mail.tau.ac.il

Maya Bechler-Speicher
Meta AI
mayabs@meta.com

Abstract

The standard in LLM-based prediction is to use the final-layer representation as the input to a downstream predictor. However, intermediate layers may encode complementary task-relevant signals. Existing approaches therefore either search for the best layer for each task or apply expensive attention-based mechanisms to learn inter-layer aggregation. In this work, we first show that such complexity is unnecessary: a lightweight Graph Neural Network over a fully connected graph of LLM layers is more efficient and achieves significantly stronger predictive performance than existing approaches. We then introduce the Cayley-Encoder, which further improves both efficiency and predictive performance by replacing the fully connected graph with a Cayley graph over $SL(2, \mathbb{Z}_n)$. These Cayley graphs provide a mathematically grounded topology that is sparse, regular by construction, and has low diameter. This enables effective communication across layers while constraining the aggregation structure and reducing the risk of GNN overfitting. In an evaluation of Cayley-Encoder across 13 tasks and 9 LLMs, Cayley-Encoder consistently outperforms baselines, achieving improvements of up to 40 percentage points in accuracy, while introducing at most 0.1% additional parameters relative to the LLM size. We further show that Cayley-Encoder is effective in few-shot regimes. Finally, we show that Cayley-Encoder outperforms LoRA fine-tuning while operating on the frozen LLM. We conclude with an explainability analysis showing that multiple layers contribute meaningfully to the final prediction, supporting our hypothesis.

1 Introduction

Large Language Models (LLMs) have become a standard backbone for learning transferable text representations across a broad range of downstream prediction tasks [Vaswani et al., 2017, Devlin et al., 2019, Brown et al., 2020]. A common efficient post-training approach is to use a frozen LLM as a feature extractor and train a lightweight prediction head on top of it. The dominant practice is to use the representation from the final layer of the LLM. This convention implicitly assumes that the last layer provides the most task-relevant representation.

A growing body of analysis shows that language models distribute information across depth. Lower layers tend to encode surface-level and lexical information, middle layers often capture syntactic

*Equal contribution.

structure, and upper layers are more strongly associated with semantic abstraction [Tenney et al., 2019, Jawahar et al., 2019, Clark et al., 2019]. Indeed, recent work has shown that intermediate layers can contain substantial task-relevant signal and may outperform final-layer representations depending on the task [Wallat et al., 2020, Fan et al., 2024, Skean et al., 2025]. There are approaches that focus on attention-based inter-layer aggregation to form a final representation based on information from all layers [ElNokrashy et al., 2024]. However, this design remains anchored to the final layer and introduces additional complexity. Other approaches modify the transformer architecture itself [Pagliardini et al., 2024, Xiao et al., 2025], but require training the architecture from scratch.

In this work, we show how inter-layer information from a frozen LLM can be aggregated in a way that is both more efficient and more effective than existing approaches, by introducing mathematically grounded geometry over the layers and formulating inter-layer aggregation as a graph learning problem. Our hypothesis is that introducing a structure that does not heavily rely on a single layer as in existing approaches is more beneficial for obtaining the final representation from the LLM. This view provides a direct mechanism for modeling interactions among layer representations without modifying the underlying LLM, in a way that is independent of the number of input tokens.

First, we show that even a simple fully connected layer graph, where every pair of layers can communicate directly through a lightweight GNN, substantially outperforms both attention mechanisms and selecting the single best-performing layer. We then show that both memory efficiency and predictive performance can be further improved by replacing the dense fully connected graph with a special sparse graph. While there are many ways to construct sparse graphs, prior work showed that GNNs tend to overfit the underlying graph structure when the structure does not carry information for the predictive task, and that regular graphs are more robust to this overfitting [Bechler-Speicher et al., 2024]. Moreover, arbitrary regular graphs may still introduce computational bottlenecks and poor communication between nodes [Alon and Yahav, 2021, Wilson et al., 2025]. To address these limitations, we introduce the Cayley-Encoder, a geometric approach that replaces the dense fully connected inter-layer graph with a sparse graph with guaranteed advantages. Cayley-Encoder maps layers to a Cayley graph over the Special Linear group $SL(2, \mathbb{Z}_n)$, which is guaranteed to be sparse, regular, and low-diameter, enabling layers to communicate effectively through the GNN.

We evaluate Fully-Connected Encoder and Cayley-Encoder on 13 diverse downstream tasks, spanning intent detection, sentiment analysis, domain classification, and semantic similarity, across 9 pretrained LLMs. Across this broad evaluation, Cayley-Encoder consistently outperforms baselines, reaching up to 40 percentage points improvement in accuracy, while adding at most 0.1% parameters relative to the base LLM. We further show that Cayley-Encoder remains effective in few-shot regimes. In addition, Cayley-Encoder outperforms LoRA fine-tuning [Hu et al., 2022] on most evaluated tasks while keeping the LLM frozen, showing that better use of existing representations can outperform direct adaptation of model weights.

Finally, we analyze the learned aggregation patterns and find that predictions rely on signals from multiple layers roughly evenly, rather than collapsing to a single dominant depth. This supports the benefit of drawing on the full depth of the LLM rather than relying on any single layer. Overall, our findings suggest that stronger representations can be obtained by allowing all layers to communicate effectively in a symmetric way.

Our main contributions are:

1. We formulate inter-layer aggregation in frozen LLMs as a graph learning problem and show that a lightweight GNN over a fully connected layer graph already substantially outperforms attention-based aggregation and best-layer optimization.

2. We introduce Cayley-Encoder, a sparse geometric inter-layer encoder based on Cayley graphs over $SL(2, \mathbb{Z}_n)$, providing a regular, sparse, and low-diameter structure for efficient communication across layers.

3. We evaluate Cayley-Encoder across 13 diverse downstream tasks and 9 LLMs, showing consistent gains over existing aggregation methods and LoRA fine-tuning while adding at most 0.1% parameters relative to the underlying LLM.

4. We show that effective inter-layer communication, rather than the identity or ordering of individual layers, is the primary driver of performance: all layers contribute roughly equally, performance is robust to layer-to-node assignment, and adding layer-position encodings does not help.

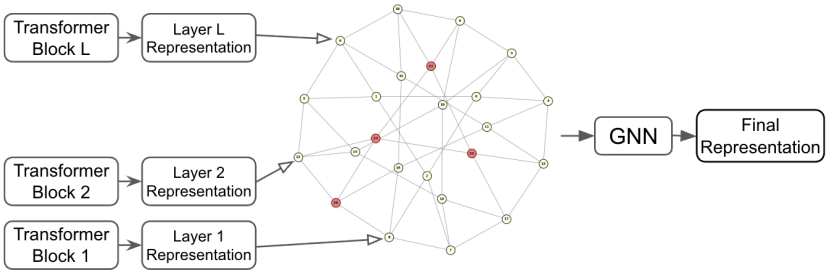


Figure 1: An overview of Cayley-Encoder. Layer representations are mapped into nodes in a Cayley Graph over the Special Linear group $SL(2, \mathbb{Z}_n)$, which is then fed into a GNN to learn the final inter-layer representation.

2 Related Work

2.1 Inter-Layer Representations in LLMs

Research on transformer depth has revealed that different layers encode different types of information. Lower layers tend to capture surface-level and lexical features, middle layers encode syntactic structure, and upper layers are associated with semantic abstraction [Jawahar et al., 2019, Tenney et al., 2019]. Importantly, intermediate layers can contain task-relevant signals that are absent from the final layer: Wallat et al. [2020] showed that intermediate layers capture factual knowledge better than the last layer, and Skean et al. [2025] demonstrated that intermediate representations consistently outperform final-layer ones across downstream tasks. These findings have motivated several approaches for aggregating information across layers. Peters et al. [2018] introduced task-specific scalar weights over BiLSTM layers in ELMo, but this restricts aggregation to a linear mixture that cannot capture non-linear inter-layer interactions. ElNokrashy et al. [2024] extended this to transformers with Depth-Wise Attention (DWAtt), which attends over layers using a query derived from the final-layer representation. While DWAtt models non-linear interactions, it anchors the aggregation to the last layer, treating earlier layers only as keys and values rather than as equal participants. Other work modifies the transformer architecture itself [Pagliardini et al., 2024, Xiao et al., 2025], but requires training from scratch. Methods such as LayerDrop [Fan et al., 2020] operate during training by randomly dropping layers, and are thus complementary to our post-training setting. Our work addresses these limitations by enabling structured inter-layer communication within a frozen LLM, without modifying the architecture or privileging any single layer (Figure 1).

2.2 Graph Neural Networks and Cayley Graphs

Graph Neural Networks (GNNs) are a family of neural networks designed to learn representations over graph-structured data. GNNs operate by iteratively propagating information between nodes through a process called message passing neural network (MPNN) [Gilmer et al., 2017]. Each MPNN layer consists of three operations: (1) each node computes messages to send to its neighbors, (2) each node aggregates the messages it receives, and (3) each node updates its representation by combining the aggregated messages with its previous state.

Many GNN variants have been proposed, differing mostly in how they aggregate information [Kipf and Welling, 2017, Xu et al., 2019]. See [Gkarmpounis et al., 2024] for a recent survey on GNNs. Bechler-Speicher et al. [2024] showed that GNNs tend to overfit the underlying graph structure when the structure does not carry information for the predictive task, and that regular graphs are more robust to this overfitting. Another limitation of GNNs is over-squashing [Alon and Yahav, 2021], where information propagating through the graph is compressed into fixed-size node representations. Deac et al. [2022] introduced Expander Graph Propagation (EGP) to improve information flow in GNNs. They address over-squashing by leveraging Cayley graphs. Cayley graphs over $SL(2, \mathbb{Z}_n)$ with the Margulis generating set are regular and sparse yet highly connected, with logarithmic diameter, allowing any two nodes to communicate in $O(\log |V|)$ steps. Cayley graphs come in fixed sizes determined by the underlying group, therefore EGP truncates nodes to match the input graph size. However, Wilson et al. [2025] showed that truncation can reintroduce bottlenecks. They therefore

proposed Cayley Graph Propagation (CGP), which instead pads the input graph with virtual nodes to preserve the complete Cayley structure. The virtual nodes are nodes used only to propagate information of the graph, but are not used for the final aggregation.

Mantri et al. [2026] introduced GLOT, which applies a GNN over a token-similarity graph constructed from the last layer's representations. Unlike our method, GLOT operates within a single layer, requires constructing the graph per input via pairwise token similarity, and produces graphs that grow linearly with input length. Our graph is fixed and determined solely by the number of LLM layers, making it constant-size regardless of input.

3 Inter-Layer Geometry

In this section, we introduce the Fully-Connected and Cayley Encoders.

We consider LLMs with L layers and an input sequence of T tokens. Each layer $\ell \in \{1, \ldots, L\}$ produces a d-dimensional hidden representation for each token $\mathbf{H}_\ell \in \mathbb{R}^{T \times d}$. We form a *layer representation* $\mathbf{z}_\ell, \ell \in \{1, \ldots, L\}$ by averaging the hidden token representation of each layer:

$$\mathbf{z}_\ell = \frac{1}{T} \sum_{t=1}^{T} \mathbf{H}_\ell[t, :] \in \mathbb{R}^d$$

Mean-pooling over tokens is a standard approach for obtaining fixed-size sentence representations from LLMs [Reimers and Gurevych, 2019]. Here, it also ensures that the resulting layer graph has a fixed size determined solely by the number of LLM layers, independent of the input sequence length. We evaluate token-level aggregation as an extension in Section 4.5.

We then map each layer representation $\mathbf{z}_\ell$ into nodes of a graph $G = (V, E)$ with nodes V and edges E. Each node v_ℓ is associated with an initial representation $h_\ell^{(0)} = \mathbf{z}_\ell \in \mathbb{R}^d$. We then apply a GNN over G to produce one final representation h_G. At layer k, the representation of node v is updated as follows:

$$m_v^{(k)} = \sum_{u \in \mathcal{N}(v)} M^{(k)} \left(h_v^{(k-1)}, h_u^{(k-1)} \right)$$

$$h_v^{(k)} = U^{(k)} \left(h_v^{(k-1)}, m_v^{(k)} \right)$$

where $m_v^{(k)}$ is the aggregated message received by node v at layer k, $\mathcal{N}(v)$ denotes the set of neighbors of node v, $M^{(k)}$ is the message function that computes the contribution from each neighbor, and $U^{(k)}$ is the update function that combines the node's previous representation with its aggregated messages. Both $M^{(k)}$ and $U^{(k)}$ are learnable functions. To obtain a graph-level representation, the final node representations are pooled; the pooling type, mean or sum, is selected as a hyperparameter.

Fully-Connected Encoder In the Fully-Connected Encoder (FC-Encoder) the layer representations are mapped to nodes in a fully connected graph, where every layer is connected to every other layer. This structure is dense, with $\binom{L}{2}$ undirected edges.

Cayley-Encoder In the Cayley-Encoder, layer representations are mapped to nodes of a Cayley Graph over $SL(2, \mathbb{Z}_n)$. A Cayley graph is a graph that describes the abstract structure of a mathematical group. Here we focus on the special linear group $SL(2, \mathbb{Z}_n)$, with elements that are defined as 2×2 matrices with integer entries modulo n and unit determinant. The resulting graph is 4-regular and has logarithmic diameter, enabling efficient global communication during message passing (see Appendix A for construction details). The number of nodes in the graph $|V_n|$ is determined by the formula:

$$|V_n| = n^3 \prod_{\text{prime } p \mid n} \left(1 - \frac{1}{p^2} \right)$$

To maintain the graph's symmetry and expansion properties, we choose the smallest n such that $|V_n| \geq L$. We map the L layer representations randomly to L nodes of the Cayley graph. When $|V_n| > L$, the remaining $|V_n| - L$ nodes are initialized as virtual nodes with zero vectors as their

representations. Following [Wilson et al., 2025], we use only the representations from the non-virtual nodes for the final hidden representation used for the prediction task.

In the next section, we demonstrate the effectiveness of Cayley-Encoder and FC-Encoder through extensive evaluation.

4 Experiments

In this section we evaluate FC-Encoder and Cayley-Encoder[2]. We evaluate on 13 diverse tasks from the MTEB benchmark [Muennighoff et al., 2023], spanning intent detection, sentiment analysis, domain classification, and semantic textual similarity. Beyond standard full-data evaluation, we assess zero-shot transfer, few-shot learning with 1–1024 samples per label, scaling behavior across LLM sizes from 14M to 2.8B parameters, comparison with LoRA fine-tuning, token-level inter-layer aggregation, and an analysis of layer importance.

Datasets. We use 13 tasks from the MTEB benchmark [Muennighoff et al., 2023]. Five are classification tasks: Banking77 (77 intent classes) [Casanueva et al., 2020], Emotion (6 emotion categories) [Saravia et al., 2018], MTOP Domain and MTOP Intent (task-oriented dialogue) [Li et al., 2021], and Poem Sentiment (4 sentiment classes) [Sheng and Uthus, 2020]. For each, we train on the training split and evaluate on the test split. The remaining 8 are semantic textual similarity (STS) tasks. We train on STSBenchmark [May, 2021] and evaluate on its test split, then perform zero-shot transfer to STS12–16, BIOSSES, and SICK-R [Soğancıoğlu et al., 2017, Agirre et al., 2012, 2013, Bandhakavi et al., 2014, Biçici, 2015, Nakov et al., 2016, Marelli et al., 2014].

Setup. We compare both encoders against a diverse set of baselines: **Last-Layer**, the last-layer representation as in [Skean et al., 2025]; **Best-Layer**, the best-performing layer selected by evaluating each layer independently; **Weighted**, a learned weighted sum of layer representations, similar to [Peters et al., 2018]; **MLP Last-Layer**, a trained MLP over the last-layer representations; **DWAtt**, a projection to 256 dimensions followed by DWAtt [ElNokrashy et al., 2024], where the projection reduces the large parameter count of applying DWAtt directly to layer representations; and **Deepset**, a Deepset over layer representations [Zaheer et al., 2017].

We evaluate on Pythia-410M (25 layers, 1024-dim) [Biderman et al., 2023], Gemma2-2B (27 layers, 2304-dim) [Team, 2024], and Llama3-8B (33 layers, 4096-dim) [AI@Meta, 2024]. All base models remain frozen during training. For the first set of experiments, we use the layer representations obtained by mean-pooling over all token representations for each layer. These pooled representations serve as the input to all methods. In this setting the input size is constant and derived from the number of layers in the LLM. We further evaluate FC-Encoder and Cayley-Encoder on all token representations across all layers. Thus making the input size dependent on samples length provided as input to the LLM.

For classification tasks, we train the encoders jointly with a linear classifier head using cross-entropy loss. For STS tasks, we encode each sentence in a pair through the frozen LLM and our encoders, compute the cosine similarity between the resulting representations, and minimize Mean Squared Error (MSE) loss with respect to the ground-truth similarity score. We use the Adam optimizer [Kingma, 2014] with learning rate and weight decay selected via Optuna [Akiba et al., 2019] hyperparameter optimization on the validation set. For FC-Encoder and Cayley-Encoder, we evaluate GIN [Xu et al., 2019] and GCN [Kipf and Welling, 2017] architectures as hyperparameters. Trainable parameters counts and full hyperparameter details are in Appendix B and C. Following the standard MTEB evaluation protocol [Muennighoff et al., 2023], we report single-run results for each configuration.

4.1 Results

Tables 1 and 2 present the evaluation. Cayley-Encoder achieves the best performance in 10 out of 15 classification configurations, all 3 supervised STS configurations, and 17 out of 21 zero-shot STS transfer configurations, making it the strongest method overall. For classification, Cayley-Encoder

[2]Code: https://github.com/tomulanovski/ILSE

Table 1: Performance comparison across 3 LLMs on 5 classification tasks. Cayley-Encoder achieves the best score in 10 out of 15 configurations, with its advantage most pronounced on larger LLMs. In all but three cases, FC-Encoder and Cayley-Encoder achieve the top two scores. We highlight in bold the best result and in blue the second best per column.

Base Model	Method	Banking77	Emotion	MTOPDomain	MTOPIntent	PoemSentiment
Pythia-410m	Last Layer	61.17	33.48	80.88	66.97	42.40
	Best Single Layer	66.67	35.02	83.78	71.18	45.67
	MLP Last Layer	83.84	33.99	96.97	83.75	75.00
	Weighted	58.50	28.26	79.79	61.68	42.60
	DWAtt	83.23	58.60	98.03	91.66	70.87
	Deepset	84.23	47.89	97.59	92.21	73.37
	FC-Encoder	**90.65**	**75.61**	98.68	**95.04**	**75.77**
	Cayley-Encoder	89.43	73.83	**98.77**	94.72	70.87
Gemma2-2B	Last Layer	62.16	26.89	80.79	68.02	35.67
	Best Single Layer	72.31	31.27	87.09	75.36	42.79
	MLP Last Layer	87.47	59.05	98.37	92.73	71.63
	Weighted	65.40	29.94	85.17	73.20	39.62
	DWAtt	88.82	61.22	98.39	90.93	67.79
	Deepset	90.03	78.77	98.92	94.37	78.56
	FC-Encoder	92.39	**79.90**	99.07	95.34	77.12
	Cayley-Encoder	**92.58**	69.60	**99.16**	**96.43**	**83.27**
Llama3-8B	Last Layer	68.25	34.23	84.42	73.39	40.96
	Best Single Layer	71.93	38.42	89.01	78.17	47.02
	MLP Last Layer	86.70	67.67	98.58	92.09	75.00
	Weighted	66.63	27.94	83.85	71.78	37.79
	DWAtt	90.04	66.55	97.97	92.41	75.00
	Deepset	87.62	71.04	98.77	95.43	77.02
	FC-Encoder	92.38	71.64	98.99	96.19	77.98
	Cayley-Encoder	**92.85**	**73.43**	**99.03**	**96.46**	**79.04**

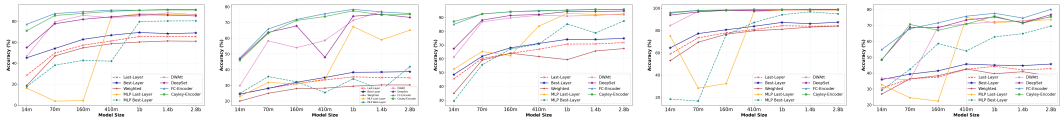
achieves average gains of 29 percentage points over Last-Layer and 24 percentage points over Best-Layer, which itself requires evaluating every layer independently. The largest gains appear on the Emotion task, with improvements of up to 40 percentage points over Last-Layer. For STS, Cayley-Encoder achieves the highest score in 17 out of 24 configurations, with the largest margins on STS13 and STS16. Comparing to multi-layer aggregation approaches, Cayley-Encoder outperforms DWAtt on all 13 tasks while using 3–5$\times$ fewer parameters (Table 7). MLP Last-Layer achieves gains on classification but provides no benefit for STS. Both encoders consistently outperform Deepset, indicating that inter-layer connectivity improves aggregation beyond permutation-invariant pooling.FC-Encoder, while on par with Cayley-Encoder in some configurations and particularly on smaller LLMs for classification, consistently outperforms all attention-based approaches, demonstrating that structured aggregation can be sufficient without attention mechanisms.

4.2 Effectiveness Across LLM Sizes

We evaluate whether the benefits of our encoders persist across LLM sizes. We conduct this evaluation on the classification tasks using the Pythia suite [Biderman et al., 2023]: 14M, 70M, 160M, 410M, 1B, 1.4B, and 2.8B parameters. Using a single model family neutralizes confounding effects due to architectural differences between LLM families. Figure 2 shows performance across Pythia sizes. Both FC-Encoder and Cayley-Encoder consistently outperform all baselines across all Pythia sizes, with gains of 20–40 percentage points over Last-Layer and Best-Layer maintained throughout. Notably, smaller LLMs equipped with our encoders can match the performance of much larger ones: Pythia-14M already achieves 95–96% accuracy on MTOP Domain, just 2–4 percentage points below Pythia-2.8B. This suggests that our approach can be particularly valuable in resource-constrained settings, extracting strong performance from small models by better utilizing their existing layer representations.

Table 2: Performance comparison across 3 LLMs on 8 STS tasks (Spearman correlation $\times 100$). Cayley-Encoder achieves the highest score in 17 out of 24 configurations. We highlight in bold the best result and in blue the second best per column.

Base Model	Method	STSBenchmark	STS12	STS13	STS14	STS15	STS16	BIOSSES	SICK-R
Pythia-410m	Last-Layer	39.12	46.96	47.00	41.45	49.32	50.37	67.30	52.55
	Best-Layer	53.53	50.62	59.27	51.61	65.59	58.02	**74.80**	**58.26**
	MLP Last-Layer	25.54	42.14	36.16	33.28	37.75	39.84	59.68	43.23
	Weighted	41.81	47.12	49.49	44.21	53.72	52.76	67.79	55.09
	DWAtt	49.40	52.51	44.57	47.46	52.51	44.80	45.29	52.43
	Deepset	39.48	52.11	42.68	43.99	48.32	40.48	44.72	44.84
	FC-Encoder	54.20	50.13	53.00	51.76	63.94	57.11	58.65	57.14
	Cayley-Encoder	**55.84**	**56.0**	**60.2**	**56.65**	**66.99**	**58.63**	56.59	55.34
Gemma2-2B	Last-Layer	36.82	33.70	45.60	37.58	50.22	51.40	58.44	43.01
	Best-Layer	52.97	43.02	58.56	52.43	65.49	58.89	**72.02**	57.24
	MLP Last-Layer	40.72	27.36	36.93	31.32	52.38	51.16	54.10	40.19
	Weighted	40.52	35.66	45.58	38.14	54.88	54.91	65.27	46.67
	DWAtt	53.37	51.86	48.14	55.29	67.73	51.11	57.08	55.36
	Deepset	36.67	43.97	38.97	35.05	50.51	47.34	31.64	42.22
	FC-Encoder	**62.86**	61.83	55.43	63.89	**74.68**	**62.96**	66.02	60.12
	Cayley-Encoder	61.62	**64.36**	**63.98**	**64.59**	73.94	62.69	64.07	**60.32**
Llama3-8B	Last-Layer	45.63	32.65	52.66	42.88	58.07	54.32	67.00	51.16
	Best-Layer	56.51	47.21	60.09	54.62	66.89	59.04	**72.83**	56.96
	MLP Last-Layer	48.91	34.88	54.80	44.08	61.19	52.55	55.35	46.62
	Weighted	45.77	32.86	59.46	54.90	67.30	58.97	52.82	56.91
	DWAtt	52.85	51.47	56.26	58.63	66.00	48.86	46.88	51.45
	Deepset	42.31	39.94	52.80	52.00	56.12	39.50	32.96	46.60
	FC-Encoder	59.33	55.19	53.51	59.88	73.69	57.25	62.81	57.51
	Cayley-Encoder	**63.05**	**65.31**	**63.53**	**70.17**	**76.96**	**63.13**	55.32	**60.74**



(a) Banking77 (b) Emotion (c) MTOP Intent (d) MTOP Domain (e) PoemSentiment

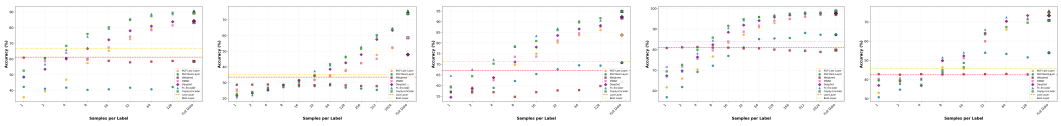
Figure 2: Performance across the Pythia models ranging from 14M to 2.8B parameters. Cayley-Encoder consistently outperforms all baselines across all model sizes, with smaller LLMs achieving performance comparable

4.3 Few-Shot Learning

We evaluate few-shot performance by restricting training to 1–1024 samples per label across all classification tasks. For Poem Sentiment, MTOP Intent, and Banking77, the available training data is limited and is fully covered at 128 samples per label. The results, presented in Figure 3, show that both encoders are highly data efficient. On Banking77, MTOP Domain, and MTOP Intent, they outperform both Last-Layer and Best-Layer baselines with as few as 8 samples per label, with Cayley-Encoder consistently achieving the highest accuracy on Banking77 and the MTOP tasks across all training set sizes. On Banking77, 32 samples per label suffice to surpass DWAtt trained on the full dataset. For Poem Sentiment, 64 samples per label enable both encoders to outperform all baselines trained with more data. For Emotion, both encoders outperform Last-Layer and Best-Layer starting from 32 samples per label, and surpass all other baselines at larger training sizes. Emotion is the only task that exhibits consistent performance improvements as training data grows; the other tasks plateau between 128 and 512 samples per label. The overall trend suggests that both encoders extract strong representations even from very few examples, making them particularly valuable for practical applications where labeled data is scarce.

4.4 Comparison with LoRA

Although our encoders are designed as lightweight post-training modules that can be combined with fine-tuning approaches such as LoRA [Hu et al., 2022], we compare them directly against LoRA to demonstrate their standalone effectiveness. LoRA inserts trainable low-rank matrices into the LLM's attention layers, modifying the model weights themselves. We searched rank $r \in \{1, 2, 3, 16\}$ via



(a) Banking77 (b) Emotion (c) MTOP Intent (d) MTOP Domain (e) PoemSentiment

Figure 3: Few-shot learning analysis using Pythia-410M. Both encoders outperform all baselines with as few as 8 samples per label on most tasks. Cayley-Encoder consistently achieves the highest accuracy on Banking77 and the MTOP tasks across all training set sizes.

Table 3: Cayley-Encoder vs. LoRA on classification tasks, measured by accuracy, and STS tasks, measured by Spearman correlation $\times 100$. Cayley-Encoder outperforms LoRA in 13 out of 15 classification and 21 out of 24 STS configurations while keeping the LLM frozen. We highlight in bold the best result per task.

Base Model	Method	Banking77	Emotion	MTOPDomain	MTOPIntent	PoemSentiment	STSBench.	STS12	STS13	STS14	STS15	STS16	BIOSSES	SICK-R
Pythia-410m	FC-Encoder	**90.65**	75.61	98.68	**95.04**	**75.77**	54.2	50.1	53.0	51.8	63.9	57.1	58.6	**57.1**
	Cayley-Encoder	89.43	73.83	**98.77**	94.72	70.87	**55.8**	**56.0**	**60.2**	**56.7**	**67.0**	**58.6**	56.6	55.3
	LoRA	82.96	**91.57**	96.64	88.25	70.96	54.1	44.6	51.7	52.3	64.8	45.8	**64.3**	52.0
Gemma2-2B	FC-Encoder	92.39	**79.90**	99.07	95.34	77.12	**62.9**	**61.8**	55.4	63.9	**74.7**	**63.0**	**66.0**	60.1
	Cayley-Encoder	**92.58**	69.60	**99.16**	**96.43**	**83.27**	61.6	60.4	**64.0**	**64.6**	73.9	62.7	64.1	**60.3**
	LoRA	56.77	70.89	89.65	80.83	61.92	58.7	49.1	57.8	60.4	69.9	51.8	58.8	54.2
Llama3-8B	FC-Encoder	92.38	71.64	98.99	96.19	77.98	59.3	55.2	53.5	59.9	73.7	57.2	62.8	57.5
	Cayley-Encoder	**92.85**	73.43	**99.03**	**96.46**	**79.04**	63.1	**65.3**	**63.5**	**70.2**	**77.0**	**63.1**	55.3	**60.7**
	LoRA	90.82	**91.98**	98.59	94.44	76.15	**65.3**	47.8	56.5	61.6	72.8	54.7	**69.6**	58.9

Optuna and report the best-performing rank per model and task (see Appendix E for details). Table 3 presents the results. For classification, at least one of our encoders outperforms LoRA in 13 out of 15 configurations. For STS, Cayley-Encoder outperforms LoRA in 21 out of 24 configurations. Notably, these gains are achieved while keeping the LLM entirely frozen and using 6–12$\times$ fewer trainable parameters than LoRA at rank 16 (Table 10). This suggests that structured aggregation over existing layer representations can be more effective than direct weight adaptation, while being substantially more parameters efficient.

4.5 Token-Level Inter-Layer Aggregation

In all experiments above, each layer is represented by a single mean-pooled vector over tokens, resulting in a graph with L nodes (one per layer). A natural extension is to operate at the token level, constructing a graph with $L \times T$ nodes,one per token per layer,allowing the GNN to model interactions across both tokens and layers simultaneously. This setting is more expressive but substantially increases the graph size, making the choice of graph topology critical for scalability. We evaluate this extension using Gemma2-2B on all classification tasks. Table 4 presents the results. FC-Encoder exceeds GPU memory on Banking77 (the largest dataset), while Cayley-Encoder handles all tasks, using 10–30$\times$ less GPU memory. This demonstrates the practical advantage of the Cayley graph's sparse $O(n)$ edge complexity over the $O(n^2)$ cost of a fully connected graph as sequences grow longer. Beyond efficiency, token-level aggregation improves accuracy. Comparing to the mean-pooled setting (Table 1), Cayley-Encoder with token-level representations achieves higher accuracy on 4 out of 5 tasks — for instance, Emotion improves from 69.60 to 89.70 and Poem Sentiment from 83.27 to 86.44. This indicates that jointly modeling information across tokens and layers captures complementary signals that are lost by mean-pooling, and suggests that token-level inter-layer aggregation is a promising direction, particularly when paired with a scalable graph topology such as the Cayley graph.

Table 4: Token-level aggregation with Gemma2-2B. Cayley-Encoder uses 10–30$\times$ less GPU memory while achieving comparable or higher accuracy. OOM: out of memory.

		Bank.	Emot.	MT-D	MT-I	Poem
Acc.	Cayley	**92.6**	89.7	99.1	**96.3**	**86.4**
	FC	OOM	**90.4**	**99.2**	93.2	80.2
GPU	Cayley	**269**	**241**	**141**	**97**	**77**
	FC	OOM	8260	2978	1583	923

4.6 Understanding Inter-Layer Communication

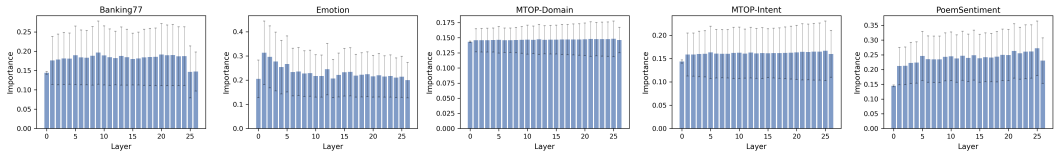


Figure 4: Per-layer contribution to the final prediction for Cayley-Encoder across classification tasks using Gemma2-2B. We estimated via GNNExplainer and averaged over all test instances. Error bars show standard deviation. All layers contribute roughly equally.

To understand how different layers contribute to the final prediction in Cayley-Encoder, and what aspects of their communication are crucial, we conduct three complementary analyses.

Table 5: Robustness to layer-to-node assignment (mean ± std over 10 random permutations).

	Bank.	Emot.	MT-D	MT-I	Poem
Pythia	90.1±.3	74.9±.8	98.7±.1	94.6±.4	71.3±1.8
Gemma	92.4±.2	73.1±1.5	99.1±.1	96.2±.1	82.3±.5
Llama	92.9±.1	72.8±1.3	98.9±.1	96.4±.2	79.2±1.3

Layer-to-node assignment does not matter. Since the Cayley graph may contain virtual nodes (when $|V_n| > L$), the specific layer-to-node assignment determines which layers are directly connected and which communicate only through virtual nodes. If the identity of communicating layers mattered, this assignment would significantly affect performance. To test this, we fix the model weights and hyperparameters for Cayley-Encoder and re-evaluate with 10 independent random layer-to-node permutations across all classification tasks and all three LLMs. Table 5 shows that accuracy remains stable with minimal variance (standard deviations below 2 percentage points in all cases), indicating that the assignment has little effect.

Layer identity is not a useful signal. We augment each node in the layer graph with a learnable positional encoding indexed by its original depth in the LLM, added to the node features prior to message passing. This provides the GNN with explicit information about which LLM layer each node represents. We train and evaluate on all classification tasks using Pythia-410M. Table 6 shows that adding positional encodings degrades performance in most cases, for both Cayley-Encoder and FC-Encoder. This suggests that layer identity is not a useful signal for the aggregation.

Table 6: Effect of adding learnable layer-position encodings. Layer identity information degrades performance in most cases.

	Bank.	Emot.	MT-D	MT-I	Poem
Cayley	89.4	73.8	98.8	94.7	70.9
Cayley (PE)	90.3	75.5	98.7	94.6	69.2
FC	90.7	75.6	98.7	95.0	75.8
FC (PE)	86.3	72.1	98.4	94.1	68.4

All layers contribute. We apply GNNExplainer [Ying et al., 2019], a post-hoc attribution method for GNNs, to Cayley-Encoder trained on each classification task with Gemma2-2B. For each test instance, GNNExplainer identifies which nodes in the layer graph the model relied on most for its prediction. Figure 4 shows the mean node importance per layer, averaged over all test instances across all five tasks. All layers contribute roughly equally, with no single layer dominating. This suggests that Cayley-Encoder aggregates complementary information from across the full depth of the LLM rather than collapsing to a single layer.

Together, these results suggest that the primary driver of Cayley-Encoder's effectiveness is enabling all layers to communicate through an expressive graph topology, rather than the specific identity or ordering of the layers. The benefit lies in structured inter-layer communication itself, not in privileging particular layers or positions.

5 Conclusion

We introduced Cayley-Encoder, a lightweight method for aggregating representations from all internal layers of a frozen LLM through a structured graph topology. Across 13 diverse tasks and 9 LLMs, Cayley-Encoder consistently outperformed all baselines by large margins, including attention-based aggregation, best-layer selection, and LoRA fine-tuning, while adding at most 0.1% parameters. These gains persist in few-shot regimes and across LLM sizes from 14M to 8B parameters. Our analysis reveals that the benefit stems from enabling effective communication across all layers, rather than from the identity or ordering of individual layers. This suggests that the internal depth of LLMs

contains complementary task-relevant signals that standard last-layer extraction fails to exploit, and that structured inter-layer aggregation provides a simple and efficient way to unlock them.

Several promising directions emerge. Scaling to larger architectures (e.g., 70B+ parameters) and generative settings remains to be explored. Additionaly, integrating structured inter-layer communication directly into LLM pre-training, rather than as a post-training module, could yield models with improved information flow across depth.

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A Cayley Graph Construction Details

The Cayley graph over $SL(2, \mathbb{Z}_n)$ is constructed using the symmetric generating set $S = \{A, A^{-1}, B, B^{-1}\}$, where

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

with all operations performed modulo n. These are the standard Margulis generators [Deac et al., 2022], producing a 4-regular graph with logarithmic diameter. Each element of $SL(2, \mathbb{Z}_n)$ corresponds to a node in the graph, and two nodes are connected if one can be obtained from the other by left-multiplication with a generator. The resulting graph is vertex-transitive, meaning all nodes are structurally equivalent.

B Trainable Parameter Counts

Table 7 reports the number of trainable parameters introduced by each method. All base LLM parameters remain frozen throughout. FC-Encoder and Cayley-Encoder introduce at most 0.1% additional parameters relative to the base model, while DWAtt requires 3–5$\times$ more parameters.

Table 7: Trainable parameters added by each method. All base model parameters remain frozen. Percentages show overhead relative to base model size.

Method	Pythia-410M	Gemma2-2B	Llama3-8B
Frozen Parameters	*410M*	*2B*	*8B*
Weighted	25	27	33
MLP	394K (0.10%)	721K (0.036%)	1.18M (0.015%)
Deepset	394K (0.10%)	721K (0.036%)	1.18M (0.015%)
FC-Encoder	395K (0.10%)	722K (0.036%)	1.18M (0.015%)
Cayley-Encoder	395K (0.10%)	722K (0.036%)	1.18M (0.015%)
DWAtt	2.0M (0.49%)	2.46M (0.12%)	3.3M (0.04%)

C Implementation Details

We provide full hyperparameter search spaces for all methods below. All methods use the Adam optimizer [Kingma, 2014] with hyperparameters selected via Optuna [Akiba et al., 2019] using MedianPruner (5 startup trials, 10 warmup epochs, checked every 5 epochs). The number of Optuna trials varied between 20 and 50 depending on the method.

FC-Encoder and Cayley-Encoder. Number of MPNN layers $\in \{1, 2\}$, dropout $\in [0.0, 0.3]$, number of MLP layers inside GIN aggregation $\in \{1, 2\}$, learning rate $\in [10^{-4}, 10^{-3}]$, weight decay $\in [10^{-4}, 10^{-3}]$. Hidden dimension is fixed to 256 and batch size to 64.

MLP Last Layer. Number of MLP layers $\in \{1, 2, 3\}$, dropout $\in [0.0, 0.3]$, learning rate $\in [10^{-4}, 10^{-3}]$, weight decay $\in [10^{-4}, 10^{-3}]$. Hidden dimension is fixed to 256.

DeepSet. Pre-pooling layers $\in \{0, 1\}$, post-pooling layers $\in \{0, 1, 2\}$, pooling type $\in \{\text{mean}, \text{sum}\}$, dropout $\in [0.0, 0.3]$, learning rate $\in [10^{-4}, 10^{-3}]$, weight decay $\in [10^{-4}, 10^{-3}]$. Hidden dimension is fixed to 256.

DWAtt. Architecture is fixed to match the original paper: bottleneck ratio $\gamma = 0.5$, positional embedding dimension $d_{\text{pos}} = 24$, and hidden dimension 256. We search dropout $\in [0.0, 0.3]$, learning rate $\in [10^{-4}, 10^{-3}]$, and weight decay $\in [10^{-4}, 10^{-3}]$.

Weighted. The architecture has no tunable structure (only L scalar layer weights with softmax). We search learning rate $\in \{10^{-4}, 10^{-3}, 10^{-2}\}$ and weight decay $\in \{0, 10^{-4}, 10^{-3}\}$.

LoRA. Rank $r \in \{1, 2, 3, 16\}$, alpha $\in \{8, 16, 32\}$, LoRA dropout $\in [0.0, 0.2]$, learning rate $\in [10^{-4}, 10^{-3}]$, weight decay $\in [10^{-4}, 10^{-3}]$. Batch size is 32 and training runs for 20 epochs.

Table 8: Hyperparameter search spaces. All methods use Optuna with MedianPruner (5 startup trials, 10 warmup steps) and 20-50 trials per study.

(a) Shared training hyperparameters

Hyperparameter	Search Space
Learning rate	{1e-4, 1e-3}
Weight decay	{1e-4, 1e-3}
Dropout	{0.0, 0.1, 0.2, 0.3}
Batch size	64
Epochs	50 (cls) / 25 (STS)

(b) Cayley-Encoder / FC-Encoder

Hyperparameter	Search Space
GNN layers	{1, 2}
GIN MLP layers	{1, 2}
Hidden dim	256
Readout	{mean, sum}

(c) MLP

Hyperparameter	Search Space
MLP layers	{1, 2, 3}
Hidden dim	256

(d) DeepSet

Hyperparameter	Search Space
Pre-pool layers	{0, 1}
Post-pool layers	{0, 1, 2}
Pooling type	{mean, sum}
Hidden dim	256

(e) DWAtt

Hyperparameter	Search Space
Bottleneck ratio	0.5
Pos. embed dim	24
Hidden dim	256

(f) Weighted

Hyperparameter	Search Space
Learning rate	{1e-4, 1e-3, 1e-2}
Weight decay	{0, 1e-4, 1e-3}

(g) LoRA

Hyperparameter	Search Space
Rank (r)	{1, 2, 3, 16}
Alpha	{8, 16, 32}
LoRA dropout	{0.0, 0.1, 0.2}
Batch size	32
Epochs	20

D Positional Encoding Ablation

We evaluate the effect of adding learnable layer-position encodings to FC-Encoder and Cayley-Encoder. Each node is assigned a learned embedding indexed by its original depth in the base LLM, added to the node features prior to message passing. We train and evaluate on both classification and STS tasks using Pythia-410M.

Table 9 presents the results. For classification, adding positional encodings yields comparable performance for Cayley-Encoder but consistently degrades FC-Encoder. For STS, positional encodings degrade Cayley-Encoder while providing marginal improvements for FC-Encoder. Overall, layer-position information does not add value, and in most cases hurts performance. This supports the finding from our main analysis that effective inter-layer communication, rather than layer identity, drives the encoders' effectiveness.

Table 9: Positional encoding (PE) ablation on Pythia-410m. Classification: accuracy (%), STS: Spearman correlation ($\times 100$). **Bold**: best per column within each encoder type.

	Classification						STS								
Method	Banking77	Emotion	MTOPDom.	MTOPInt.	PoemSent.	Avg	STS-B	STS12	STS13	STS14	STS15	STS16	BIOSSES	SICK-R	Avg
Cayley-Encoder	89.43	73.83	**98.77**	**94.72**	**70.87**	85.52	**55.8**	**56.0**	**60.2**	**56.7**	**67.0**	**58.6**	**56.6**	**55.3**	**58.3**
Cayley-Encoder (PE)	**90.25**	**75.52**	98.72	94.58	69.23	**85.66**	55.3	51.4	54.8	48.2	63.0	53.2	44.8	54.2	53.1
FC-Encoder	**90.65**	**75.61**	**98.68**	**95.04**	**75.77**	**87.15**	54.2	50.1	53.0	51.8	63.9	**57.1**	**58.6**	**57.1**	55.7
FC-Encoder (PE)	86.33	72.07	98.36	94.10	68.37	83.85	**55.3**	**52.5**	**55.9**	**54.2**	**65.0**	56.3	56.5	56.6	**56.5**

E Additional LoRA Results

Table 10 compares the trainable parameter counts of LoRA at ranks 3 and 16 against Cayley-Encoder. At rank 16, LoRA introduces 6–12× more trainable parameters than Cayley-Encoder, depending on the model size. Despite this parameter advantage, Cayley-Encoder outperforms LoRA on the majority of tasks while keeping the LLM entirely frozen.

Table 10: Trainable parameters for LoRA (ranks 3 and 16) compared to Cayley-Encoder. Percentages show overhead relative to base model size.

Method	Pythia-410M	Gemma2-2B	Llama3-8B
Frozen Parameters	*410M*	*2B*	*8B*
LoRA ($r = 3$)	442K (0.11%)	1.20M (0.046%)	2.56M (0.032%)
LoRA ($r = 16$)	2.36M (0.58%)	6.39M (0.25%)	13.6M (0.17%)
Cayley-Encoder	395K (0.10%)	722K (0.036%)	1.18M (0.015%)

F Cayley Graph vs. Random Regular Graph

To determine whether the performance of Cayley-Encoder stems from its algebraic structure or simply from regularity and sparsity, we replace the Cayley graph with a random 4-regular graph. The random graph matches the Cayley graph in degree, regularity, and sparsity, but lacks its algebraic expansion properties. We evaluate across Pythia-410M, Gemma2-2B, and Llama3-8B on all classification and STS tasks, with hyperparameters tuned independently for each method.

Tables 11 and 12 present the results. The Cayley graph outperforms the random 4-regular baseline on 9 out of 15 classification and 22 out of 24 STS task–model pairs, with the largest gains on Llama3-8B. This indicates that the algebraic structure of the Cayley graph — not merely its regularity or sparsity — contributes meaningfully to performance.

Table 11: Ablation isolating the contribution of the Cayley-Encoder-graph structure. Our Cayley-Encoder-Encoder uses a 4-regular topology; we ask whether its gains over FC-Encoder come from sparsity alone or from the specific Cayley-Encoder connectivity. Replacing the Cayley-Encoder graph with a random 4-regular graph in classification tasks shows comparable results. Cayley-Encoder-Encoder is better in 9 out of 15 cases. **Bold**: best per column.

Base Model	Graph Type	Banking77	Emotion	MTOPDomain	MTOPIntent	PoemSentiment
Pythia-410m	Random-Regular	**90.38**	**75.15**	**98.80**	**94.73**	70.77
	Cayley-Encoder	89.43	73.83	98.77	94.72	**70.87**
Gemma2-2B	Random-Regular	92.16	**76.50**	99.12	96.21	82.31
	Cayley-Encoder	**92.58**	69.60	**99.16**	**96.43**	**83.27**
Llama3-8B	Random-Regular	92.23	72.99	98.90	96.29	**79.81**
	Cayley-Encoder	**92.85**	**73.43**	**99.03**	**96.46**	79.04

Table 12: Ablation isolating the contribution of the Cayley-graph structure. Our Cayley-Encoder uses a 4-regular topology; we ask whether its gains over FC-Encoder come from sparsity alone or from the specific Cayley connectivity. Replacing the Cayley graph with a random 4-regular graph degrades performance on 22 of 24 task–model pairs across 3 LLMs and 8 STS benchmarks, showing that the structured connectivity – not the regularity itself – drives the improvement. **Bold**: best per column.

Base Model	Method	STSBenchmark	STS12	STS13	STS14	STS15	STS16	BIOSSES	SICK-R
Pythia-410m	Random-Regular	53.19	51.69	56.81	52.82	62.19	54.27	47.81	**55.66**
	Cayley-Graph	**55.84**	**55.97**	**60.25**	**56.65**	**66.99**	**58.63**	**56.59**	55.34
Gemma2-2B	Random-Regular	59.85	52.57	58.25	58.73	72.40	61.11	62.62	58.34
	Cayley-Graph	**61.62**	**64.36**	**63.98**	**64.59**	**73.94**	**62.69**	**64.07**	**60.32**
Llama3-8B	Random-Regular	54.46	44.55	48.92	46.02	65.47	55.55	**57.68**	53.51
	Cayley-Graph	**63.05**	**65.31**	**63.53**	**70.17**	**76.96**	**63.13**	55.32	**60.74**

G Hardware Details

All experiments were conducted on an HPC cluster with access to NVIDIA A6000, A100, V100, and GeForce RTX 3090 GPUs. We do not report training time comparisons, as experiments were distributed across different GPU types and scheduling queues, making direct timing comparisons unreliable.

H Asset Licenses and Terms of Use

All models and datasets used in this work are publicly available and used in accordance with their respective licenses.

H.1 Model Licenses

The following large language models served as the frozen backbones for our experiments [Biderman et al., 2023, Team, 2024, AI@Meta, 2024] :

- **Pythia Suite (14M–2.8B)**: Licensed under the **Apache License 2.0**.
- **Gemma2-2B**: Released under the **Gemma Community License.** .
- **Llama3-8B**: Released under the **Meta Llama 3 Community License Agreement**.

H.2 Dataset Licenses (MTEB Benchmark)

The datasets used for evaluation were accessed via the Massive Text Embedding Benchmark (MTEB) [Muennighoff et al., 2023] which is licensed under the **Apache License 2.0**. Individual dataset licenses are summarized below:

Task Category	Dataset	License
Classification	Banking77	**CC BY 4.0**
	Emotion	**CC BY 4.0**
	MTOP (Domain/Intent)	**CC BY-SA 4.0**
	Poem Sentiment	**Apache License 2.0**
STS	STSBenchmark	**CC BY-SA 4.0**
	STS12–16	**CC BY-SA 4.0**
	BIOSSES	**CC BY-NC 4.0**
	SICK-R	**CC BY-NC-SA 4.0**

Table 13: Verified licenses for the models and datasets used in this work.